Layout experiment 1.a: design review (session 2.b)

Board 49 × 68 mm (session 2.b, Nick's 3 + 3 mm), the fitted 20 W inductors' courtyards + 2 mm as the keep-out, U1 in the west pocket 8.8 mm from the edge (the centre placement did not route: OPTIONS.md addendum A3), the SPI on a reserved Internal 2 lane routed first, the inserts 1.5 mm nearer the wall, the jumpers on top at the north edge, JLCPCB's limits in the design rules (DFM.md), no router via in any solder pad. Left: Sofar's mote, its copper placed as this board places the block. Right: this board. Generated by `tools/review_views.py`; the numbers are in `out/m4/summary.json`. The before / after page shows session 2's board beside this one, layer by layer.

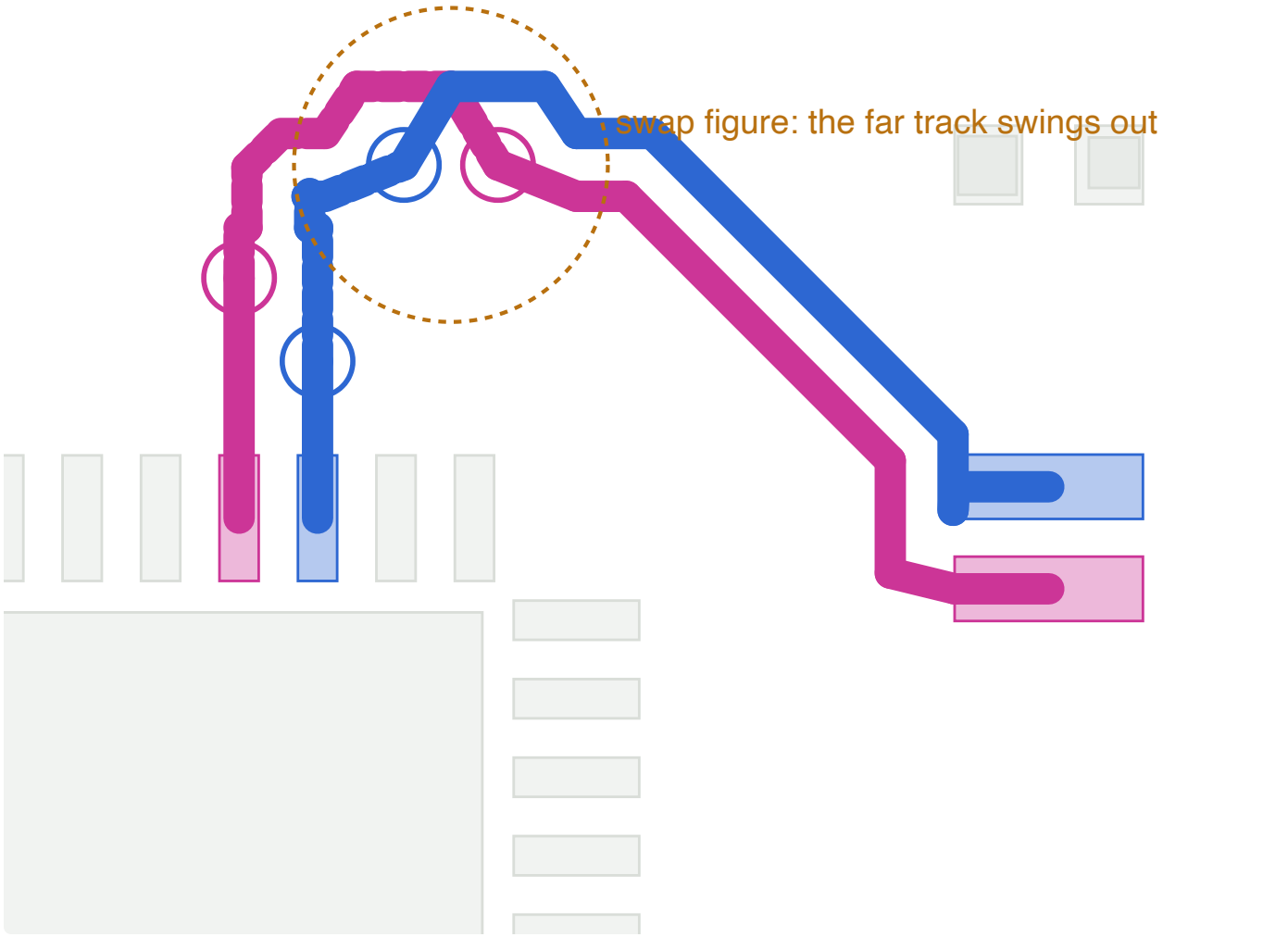
Colours: data pairs bus VBUS 3V3 rails 5 V to the Pi signals / GND tracks GND fills. Left: Sofar's mote, its copper placed as this board places the block. Right: this board. Same renderer, same region, same scale (grid 0.1 mm cells; the images are 40 px/mm).

Critical pairs: P versus N length

The two 10BASE-T1L data pairs are the board's only length-critical nets (BRIEF §6: each leg no longer than the mote's; the mote's own legs differ too: Sofar's port-1 pair is 9.5 / 9.5 by the limit but his copper is not equal either). Track lengths only, vias not counted, from `tools/pair_lengths.py` walking each net from U1's pin to the transformer pad. Where the P–N difference comes from, per pair:

Port 1: BM1_DATA_P 9.39 mm vs BM1_DATA_N 8.77 mm (difference +0.62 mm; limit 9.5 mm each)

Part of the route	P mm	N mm	P – N	Why
pins' exit (stub + start vias)	1.85	1.85	+0.00	two 0.45 mm vias do not fit side by side at U1's 0.5 mm pin pitch, so one leg's via sits 0.53 mm further out and that leg runs longer at the pins' pitch before narrowing to the pair's 0.4 mm (port 1); on Top both stubs are equal (port 2)
swap figure	2.68	2.49	+0.19	the pin order and the pad order are opposite, so P and N must cross once: the near track dives and the far track swings 0.5 mm out around its via, which adds ≈ 0.5 mm to the far track
runs and bends	3.12	3.19	-0.07	on a coupled pair every bend puts one leg on the inside and the other on the outside: ≈ 0.4 mm per 90° turn, ≈ 0.17 mm per 45°; the bends do not all cancel
approach into the pad	1.74	1.23	+0.51	the last 1.4 mm: the legs spread from the 0.4 mm pitch to the pads' 0.65 mm pitch and enter straight; equal by construction
total	**9.39**	**8.77**	**+0.62**	at 7.5 MBaud (10BASE-T1L) the skew of 0.6 mm is ≈ 4 ps against a 133 ns symbol; the brief holds each leg to the mote's length, not P to N

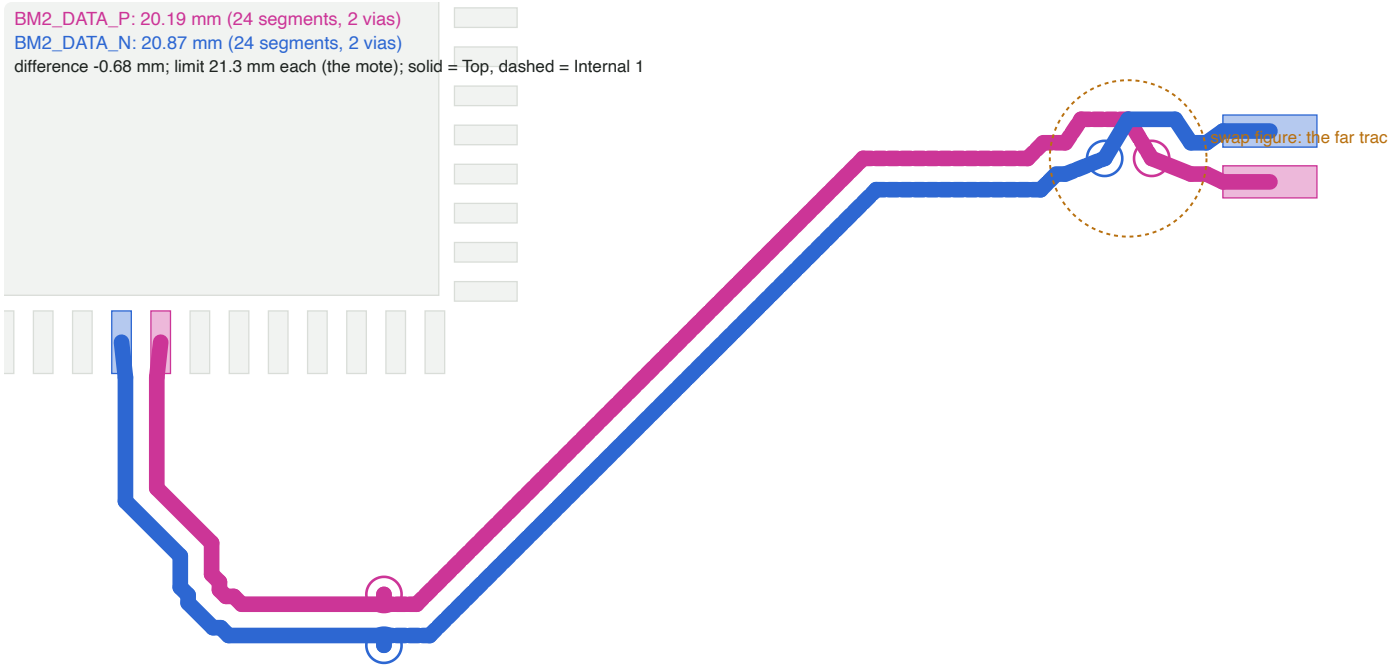


Port 1: both legs from U1's pins to the transformer pads; P magenta, N blue; solid Top, dashed Internal 1; circles are vias; the dashed ring marks the swap figure.

Port 2: BM2_DATA_P 20.19 mm vs BM2_DATA_N 20.87 mm (difference -0.68 mm; limit 21.3 mm each)

Part of the route	P mm	N mm	P – N	Why
pins' exit (stub + start vias)	1.87	2.04	-0.17	two 0.45 mm vias do not fit side by side at U1's 0.5 mm pin pitch, so one leg's via sits 0.53 mm further out and that leg runs longer at the pins' pitch before narrowing to the pair's 0.4 mm (port 1); on Top both stubs are equal (port 2)
swap figure	2.57	2.40	+0.17	the pin order and the pad order are opposite, so P and N must cross once: the near track dives and the far track swings 0.5 mm out around its via, which adds ≈ 0.5 mm to the far track

runs and bends	14.92	15.58	-0.66	on a coupled pair every bend puts one leg on the inside and the other on the outside: ≈ 0.4 mm per 90° turn, ≈ 0.17 mm per 45°; the bends do not all cancel
approach into the pad	0.83	0.86	-0.03	the last 1.4 mm: the legs spread from the 0.4 mm pitch to the pads' 0.65 mm pitch and enter straight; equal by construction
total	**20.19**	**20.87**	**-0.68**	at 7.5 MBaud (10BASE-T1L) the skew of 0.6 mm is ≈ 4 ps against a 133 ns symbol; the brief holds each leg to the mote's length, not P to N



Port 2: both legs from U1's pins to the transformer pads; P magenta, N blue; solid Top, dashed Internal 1; circles are vias; the dashed ring marks the swap figure.

To equalise P and N the next router step would mirror the start stagger at the pads (the far-via leg gets the near pad) and place the swap where a bend cancels it; both are emitter changes, not placement.

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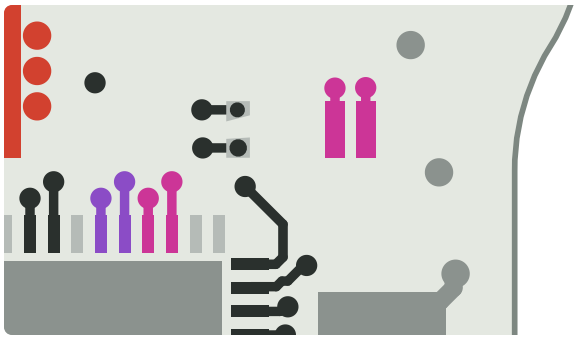
port1 pair x -3.5…10.5, y 25.5…32.5 mm · block ADIN

Port-1 data pair: U1 pins 27/28 (north edge of U1 at rot 90) to T1 pads 1/2 in the band; the mote's U1 and T1 sit elsewhere, so the mote view shows only the ADIN block's copper placed as here

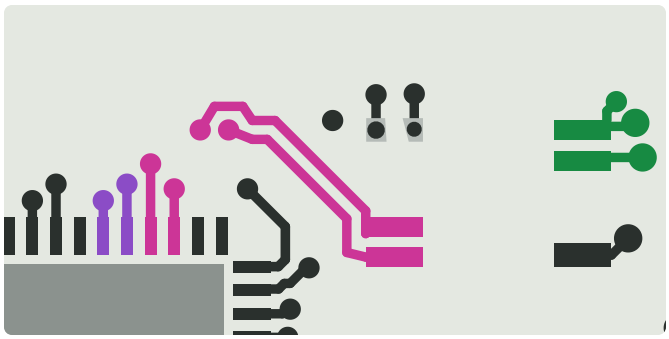
> **Fixed / as built:** Session 2.b: the pair's 45° runs are single segments (15 segments per leg, was 25) and the P leg keeps 0.11 mm under the mote's 9.5 (9.39 / 8.77; T1 0.2 mm north of session 2's relation). Two vias straight out of U1's pins, Internal 1, one swap, Top into T1's pads along the straight lane.

> **Versus the mote:** Sofar leaves the pins by vias 0.75 / 1.1 mm out and runs the port on Internal 1 too, but enters T1's pads from the body side with a via behind each pad; here the pads face U1 and the pair enters from the open side.

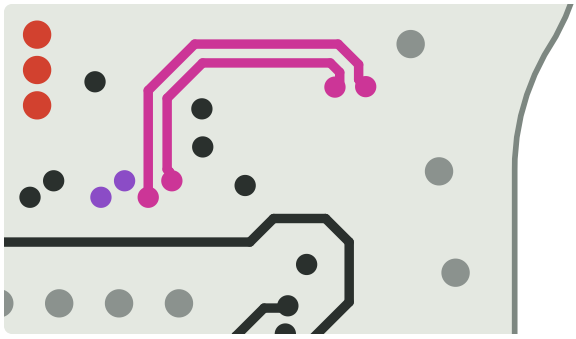
> **Still open:** 0.11 mm of margin on P: a rerun is not byte-identical. The centre placement (OPTIONS addendum A3) would give the mote's own geometry but needs the body-side entry to route.



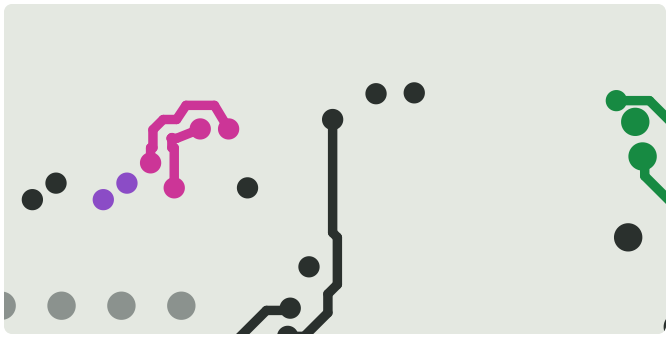
Sofar's mote, Top (block ADIN transform)



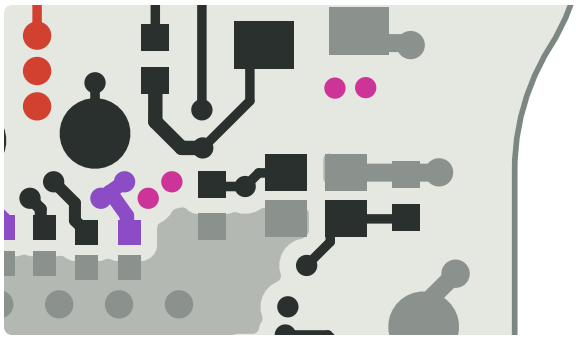
This board, Top



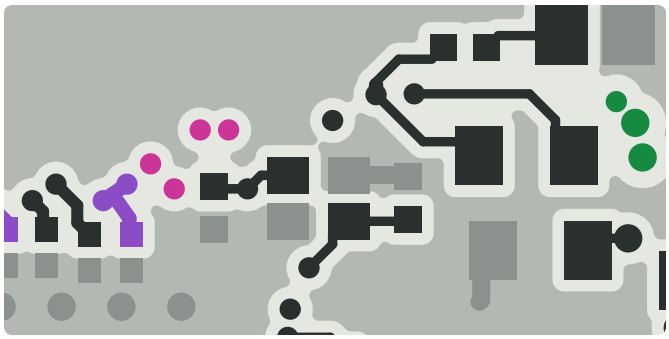
Sofar's mote, Internal 1 (block ADIN transform)



This board, Internal 1



Sofar's mote, Bottom (block ADIN transform)



This board, Bottom

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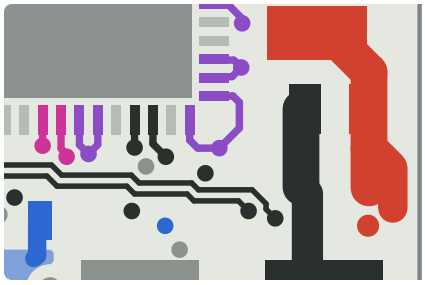
port2 pair x -4.0…14.0, y 34.0…41.5 mm · block ADIN

Port-2 data pair: U1 pins 4/5 (south edge) through the pocket's south exit to T2 pads 1/2

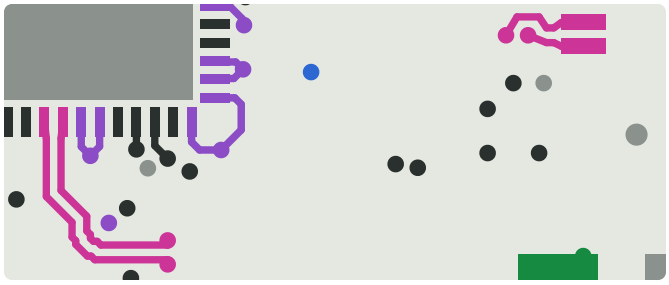
> **Fixed / as built:** Leaves U1 pins 4/5 on Top through the pocket's south exit (3 mm taller than session 2), swaps once, gets its via pair beside T2's approach. 19.93 / 20.61 mm (mote 21.3; session 2 17.98 / 18.62: the exit is 1.25 mm further south now).

Versus the mote: Sofar's port 2 is 21 mm of Internal 1 to a T2 on the far side of the processor.

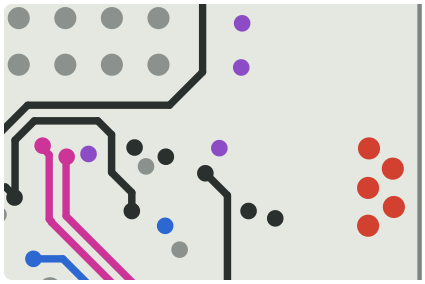
Still open: None at the bar; the margin is 0.7 mm.



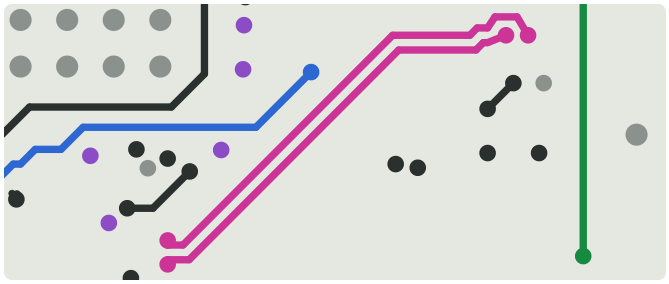
Sofar's mote, Top (block ADIN transform)



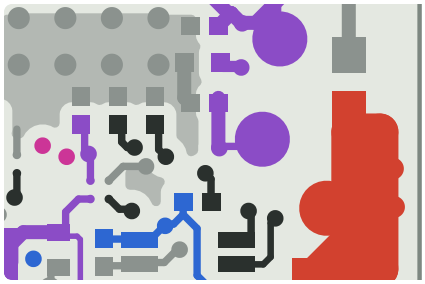
This board, Top



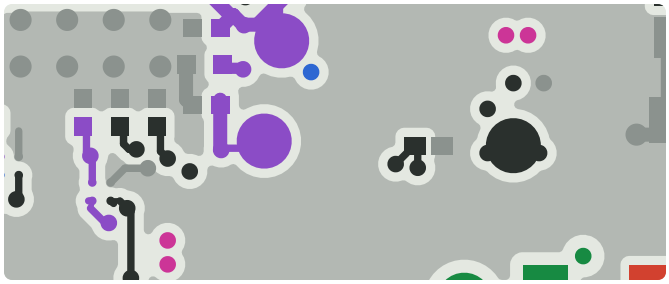
Sofar's mote, Internal 1 (block ADIN transform)



This board, Internal 1



Sofar's mote, Bottom (block ADIN transform)



This board, Bottom

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spi lane x -7.5…29.5, y 27.0…38.5 mm · no mote counterpart

The SPI lane: from Sofar's vias at U1's west side east on Internal 2 under U1 and across the band to J1 pins 19-24, routed first (session 2: 25 mm down the west edge past the inserts)

Fixed / as built: Nick's rule for 2.b: the SPI (SCK, MOSI, MISO, ~CS) is routed FIRST on a reserved Internal 2 lane from Sofar's fan-out vias at U1's west side, east under U1 and across the band to J1 pins 19-24; Bottom is forbidden for it and Top serves only the pin escape. Session 2's 25 mm detour down the west edge past the inserts is gone: 34 / 35 / 40 / 41 mm of new track against 66.5 / 67.6.

Versus the mote: No counterpart: the mote's SPI goes to its own processor on Internal 1.

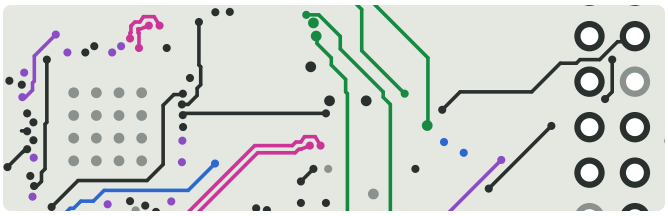
Still open: The lane crosses the port-1 bus legs (Internal 1) and the 3V3 / 1V8 corridors once each. Length matching is not required (LESSONS §5).

no mote counterpart on Internal 2 (In3)



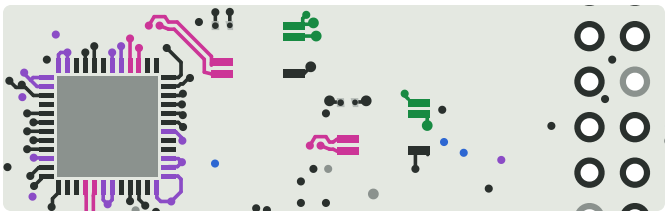
This board, Internal 2 (In3)

no mote counterpart on Internal 1



This board, Internal 1

no mote counterpart on Top



This board, Top

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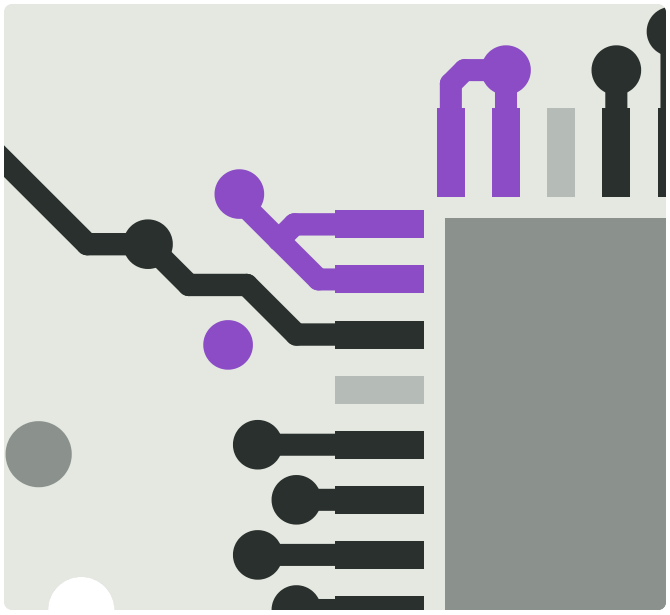
int fanout x -8.5…-2.5, y 29.0…34.5 mm · block ADIN

~{ADIN_INT} fan-out west of U1: U1.39 -> Sofar's own via (2.8 mm inside the edge now that U1 is 3 mm east: no hand via) -> R1.1

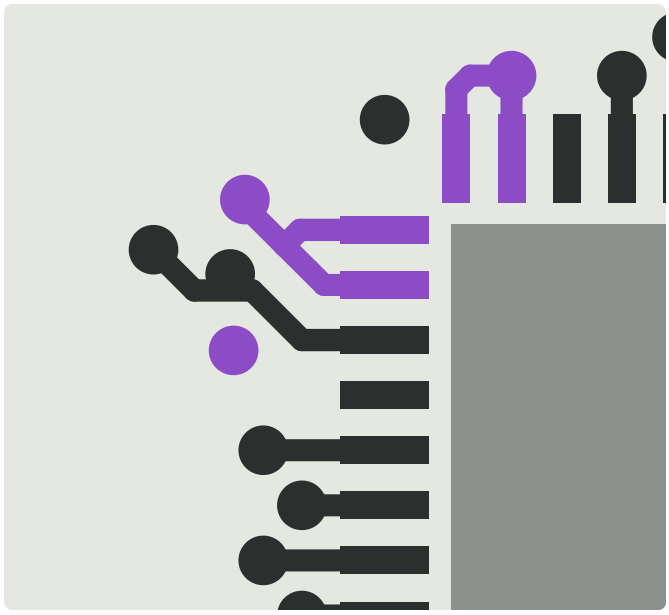
> **Fixed / as built:** Sofar's ~{ADIN_INT} fan-out whole and Sofar's own: with U1 3 mm further from the west edge (Nick, 2026-10-08) the mote's via lands 2.8 mm inside the board, so the ADIN region is the mote's full one again and the hand via of sessions 2 / 2.b is gone (the mechanism stays in m3_copy for a layout that needs it).

> **Versus the mote:** Identical copper.

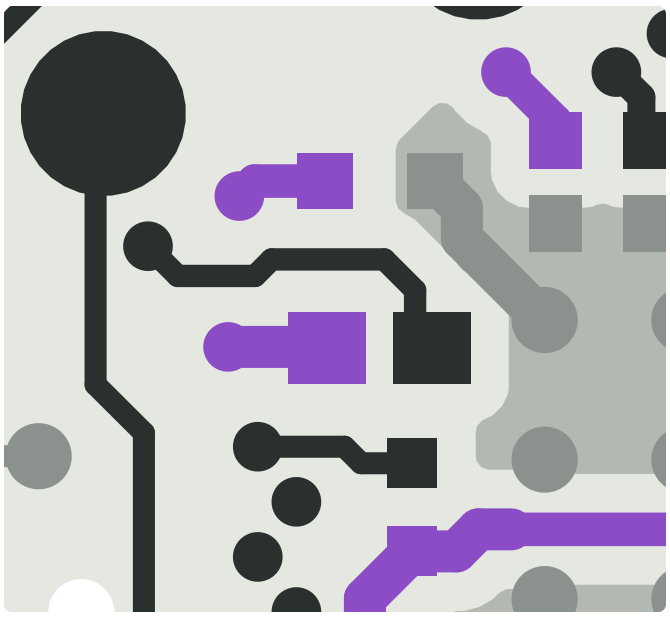
> **Still open:** None.



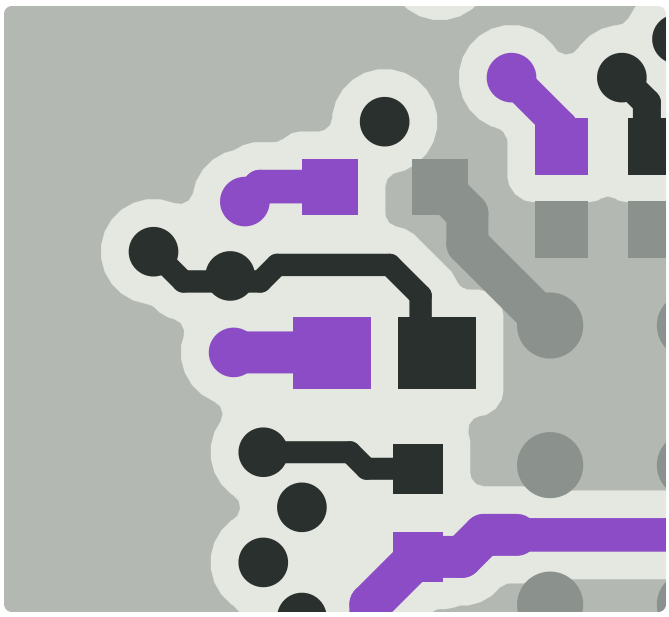
Sofar's mote, Top (block ADIN transform)



This board, Top



Sofar's mote, Bottom (block ADIN transform)



This board, Bottom

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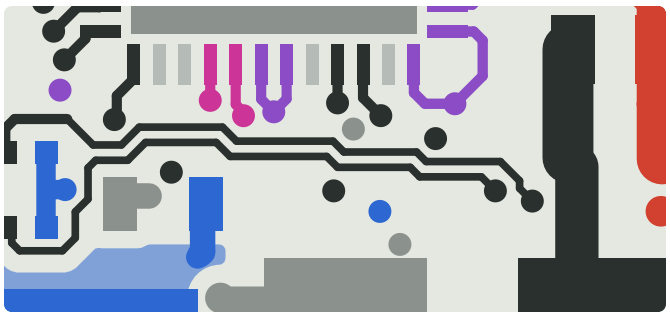
u2 u3 pocket exit x -7.0…6.0, y 36.0…42.0 mm · block ADIN

The pocket's south exit: U2 / U3 (1.8 V and 3.3 V switches, bottom) with their fan-out vias, the port-2 pair above, 3V3 / 1V8 / ADIN_PWR leaving east; 3 mm taller than session 2

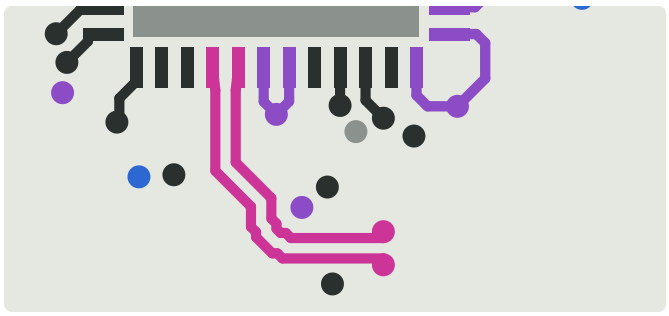
> **Fixed / as built:** The exit is 3 mm taller (MP1 2.5 mm south, the pocket y 26.2 … 41.3) and U1 is 3 mm east of the insert column; U2 / U3's balls get their vias before the pairs (fan-out first) anywhere but the pairs' exit lanes and never in a solder pad (QE round 3 F1); 3V3, 1V8 and ADIN_PWR leave east on Internal 2 under the band.

> **Versus the mote:** Sofar has U2 / U3 beside the processor with room on three sides.

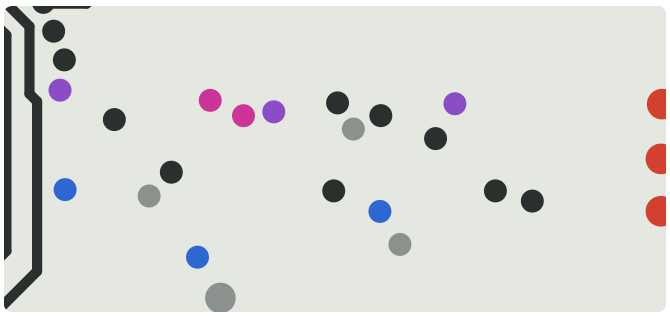
> **Still open:** U2.B2 (ADIN_PWR) and U3.A2 (3V3) still found no fan-out spot within 1.7 mm and were routed afterwards (connected).



Sofar's mote, Top (block ADIN transform)



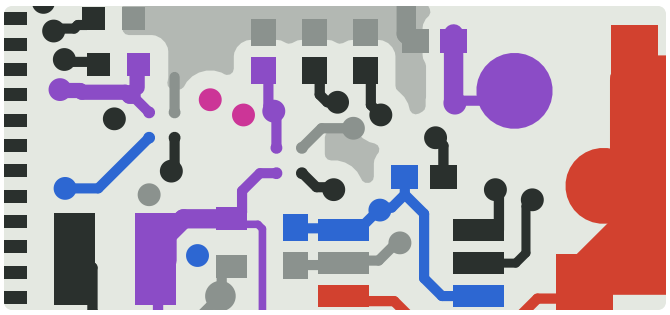
This board, Top



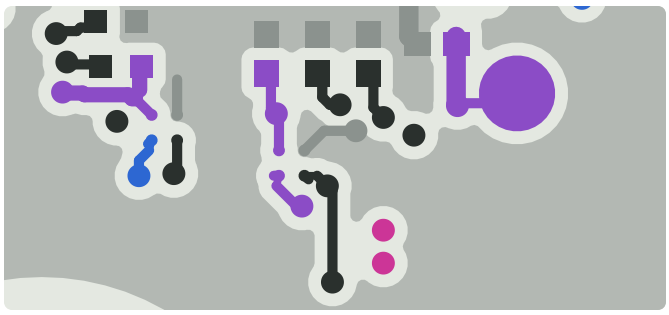
Sofar's mote, Internal 2 (In3) (block ADIN transform)



This board, Internal 2 (In3)



Sofar's mote, Bottom (block ADIN transform)



This board, Bottom

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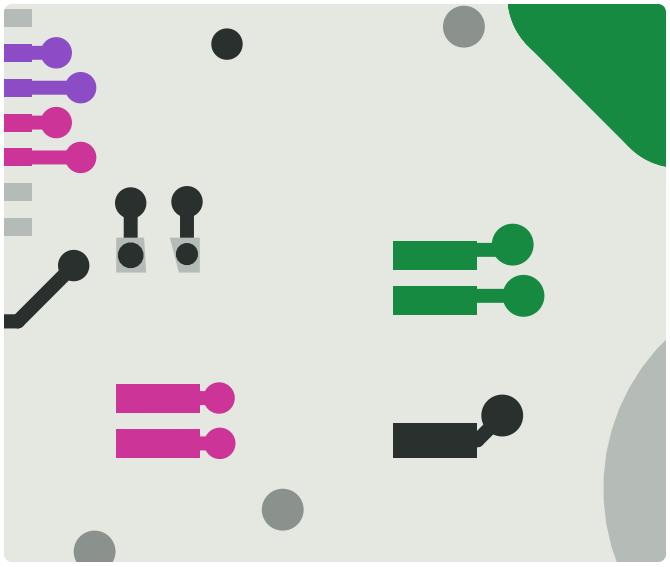
t1 cluster x 2.5…12.0, y 24.5…32.5 mm · block P1T

T1 cluster (P1T block) and the bus legs to L1

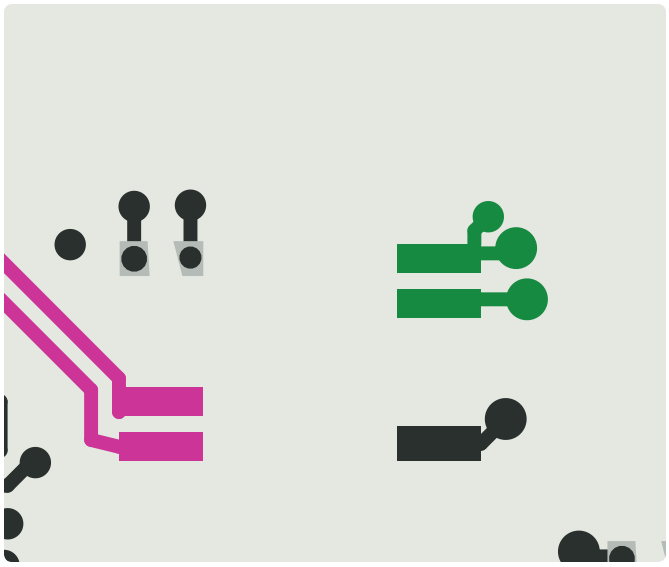
> **Fixed / as built:** T1 at (6.7, 29.45): session 2's relation to U1 but 0.2 mm north (the pair's margin), the group 3 mm east with U1. Rigid copy; C19 / R14 as their own rigid piece at the mote's spot relative to T1.

> **Versus the mote:** The T1 cluster is a rigid copy (pads, Sofar's tracks and vias to 0.001 mm); only the data-pad approach is new.

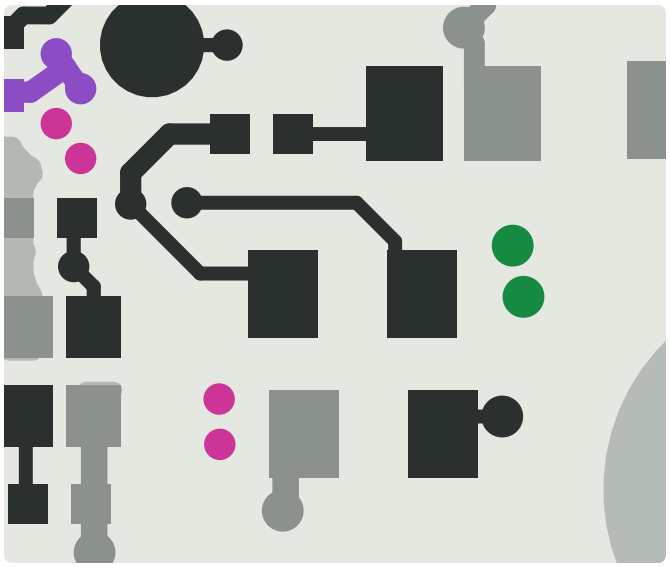
> **Still open:** Sofar's pads 6/7 are 0.24 mm apart (the bus class asks 0.35): 4 DRC items per transformer, in the exclusion table as on the mote.



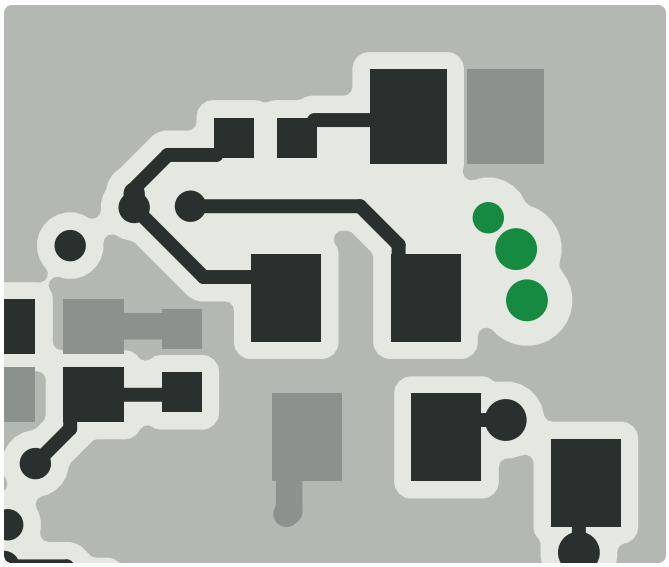
Sofar's mote, Top (block P1T transform)



This board, Top



Sofar's mote, Bottom (block P1T transform)



This board, Bottom

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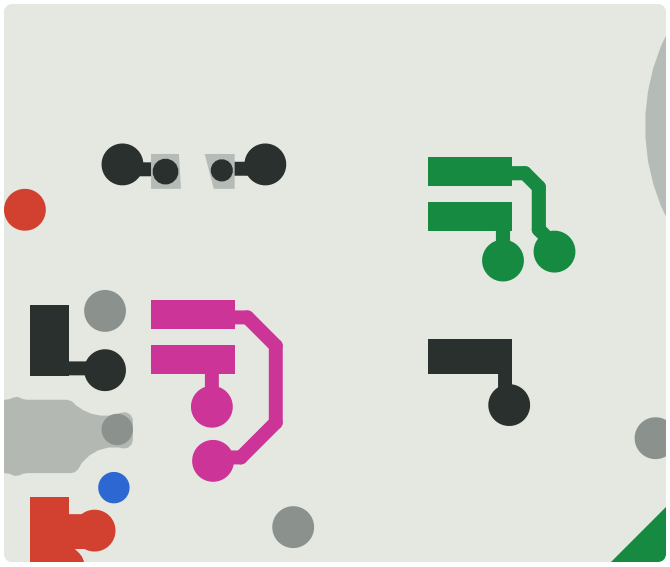
t2 cluster x 9.0…18.5, y 30.0…38.0 mm · block P2T

T2 cluster (P2T block) east of T1

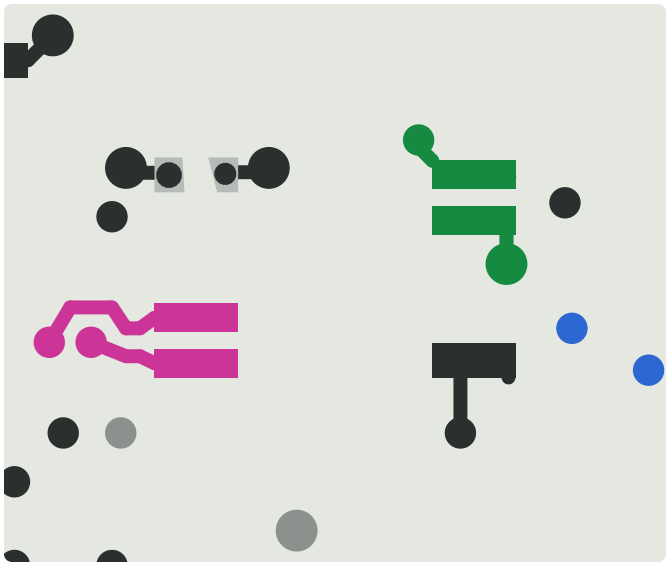
> **Fixed / as built:** T2 at (13.7, 33.75), session 2's relation to U1; the P2T region ends at mote x 138.6 as session 2.

> **Versus the mote:** Everything in the cluster is the mote's copper; the dropped GND via's job is done by this board's own stitching vias.

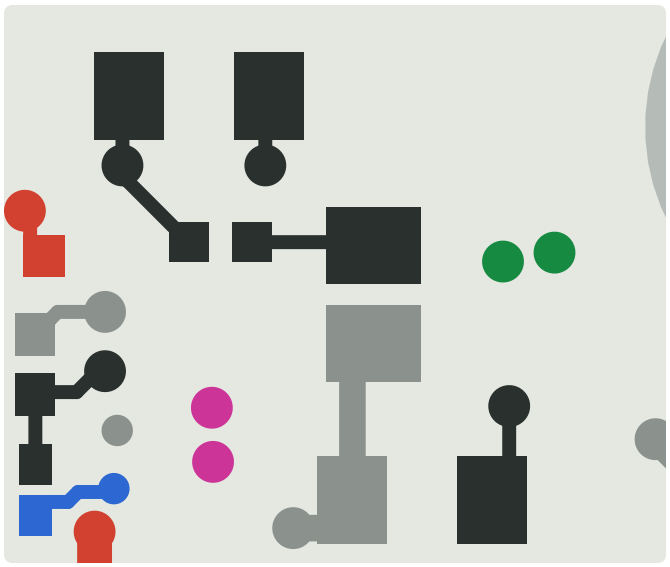
> **Still open:** None.



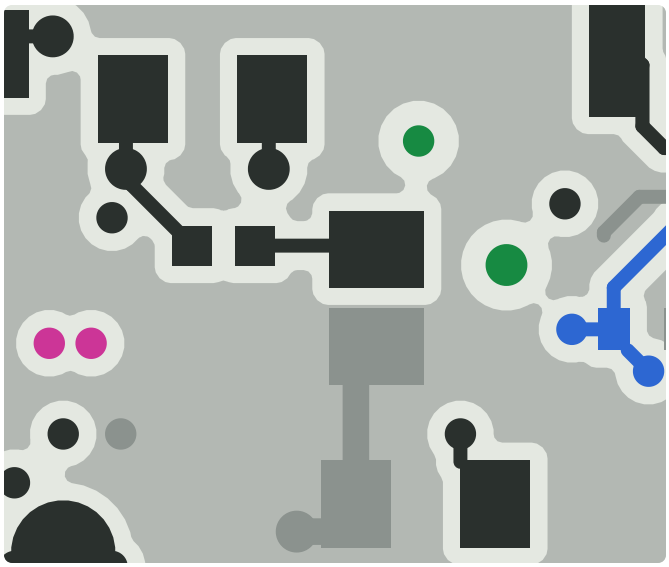
Sofar's mote, Top (block P2T transform)



This board, Top



Sofar's mote, Bottom (block P2T transform)



This board, Bottom

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north band x 4.5…24.5, y -0.5…8.0 mm · no mote counterpart

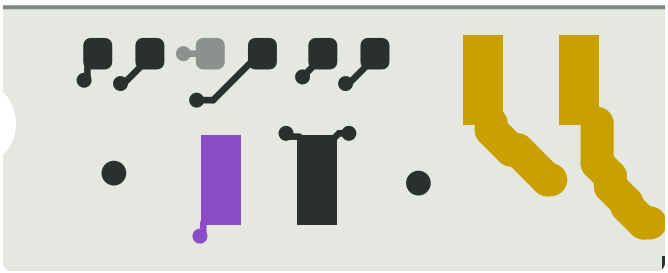
North band: the LEDs D10 / D8 / D9 on the edge, JP1 (5 V to the Pi) and JP2 (LED supply) on top beside them, FID3 / FID4; the LED resistors and FID1 / FID2 underneath

> **Fixed / as built:** Both cut jumpers on top at the north edge with nothing tall beside them (Nick): JP1 (5 V to the Pi) at the band's east end, JP2 (LED supply) behind the LEDs; FID3 / FID4 ≥ 3.35 mm from the edge (JLCPCB, DFM row 26). The 5 V path L6 → JP1 → J1 pins 2/4 crosses J1's north end on an inner layer.

> **Versus the mote:** No counterpart (the LEDs, the jumpers and the Pi are new on this board).

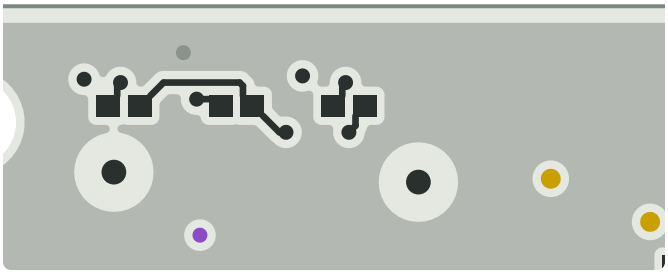
> **Still open:** The 5 V path is ≈ 33 mm of 1.0 mm track instead of session 2's 8 (OPTIONS addendum A4: ≈ 35 mV at 1 A).

no mote counterpart on Top



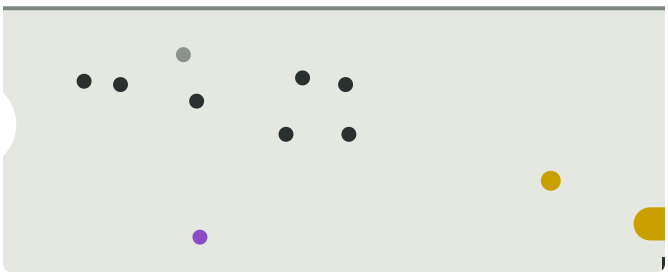
This board, Top

no mote counterpart on Bottom



This board, Bottom

no mote counterpart on Internal 1



This board, Internal 1

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u11 payload x 27.0…38.5, y 49.0…62.0 mm · no mote counterpart

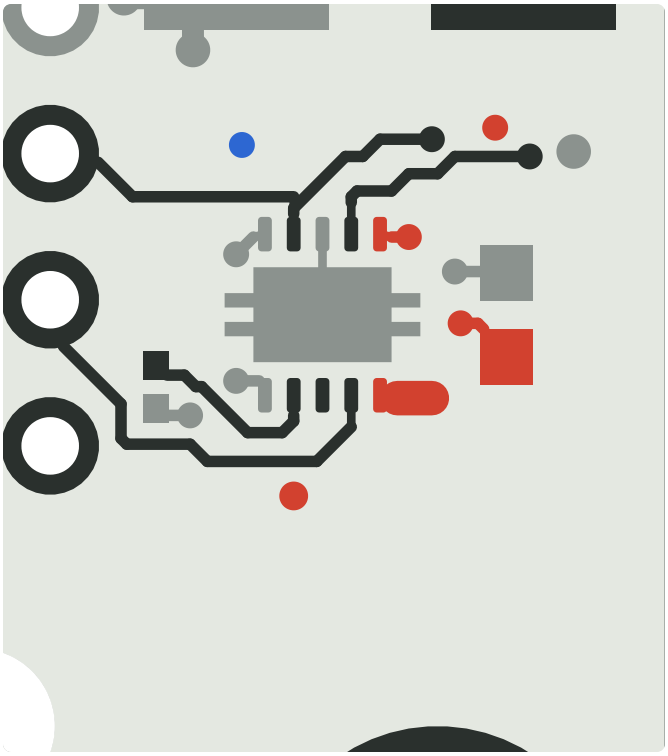
Load switch U11 at the strip's south end: VBUS in (pre-stitched via), VBUS_OUT to J5 (0.6 mm), its signals to J1

> **Fixed / as built:** Unchanged from session 2 but for the frame: VBUS pre-stitched at U11 pin 1, VBUS_OUT (0.6 mm) on Internal 2 to J5 3 mm further south.

> **Versus the mote:** No mote counterpart (the load switch is new on this board).

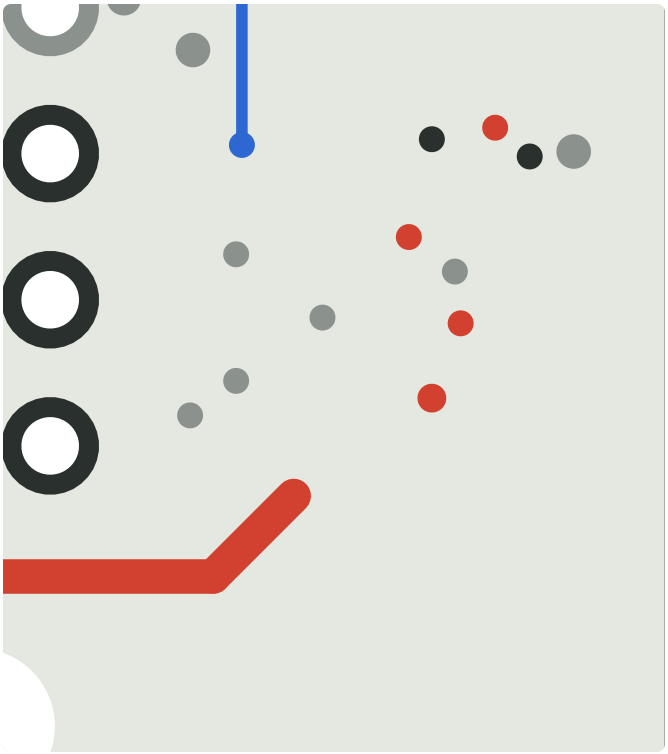
> **Still open:** None.

no mote counterpart on Top



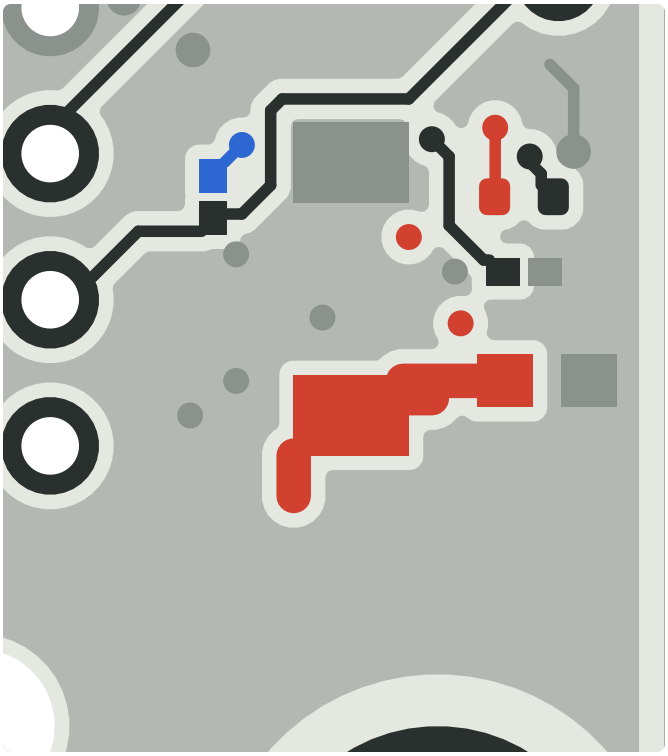
This board, Top

no mote counterpart on Internal 2 (In3)



This board, Internal 2 (In3)

no mote counterpart on Bottom



This board, Bottom

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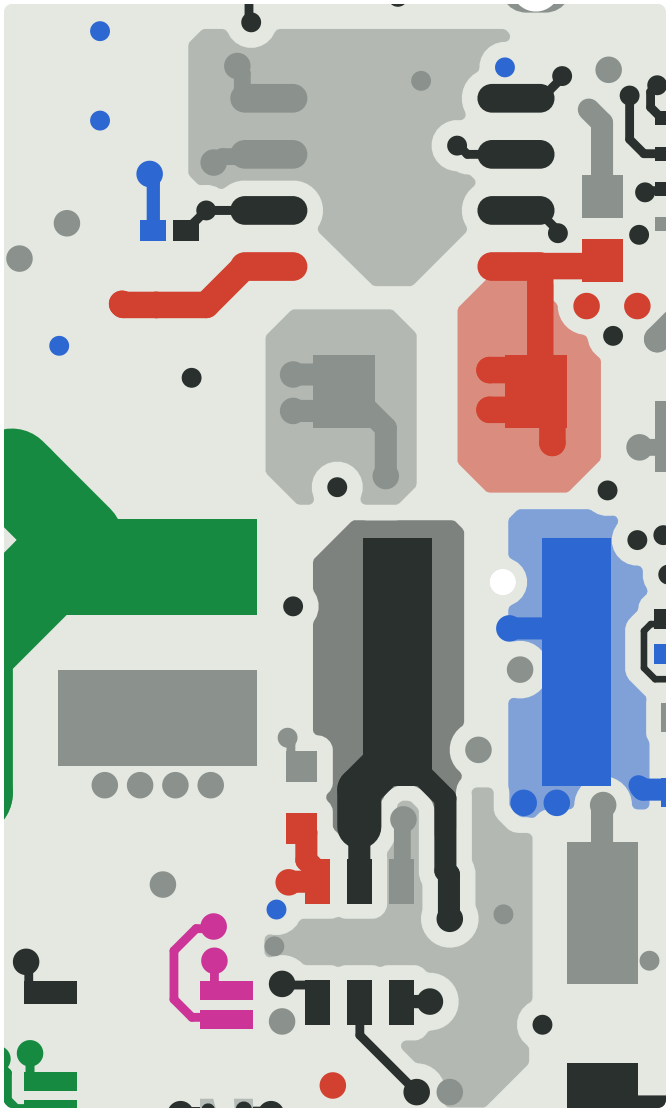
pi5v strip north x 23.5…38.5, y 0.0…25.0 mm · block B5V

5 V cell at the strip's north end and the 5 V path: L6 -> JP1 in the north band -> J1 pins 2/4 across J1's north end

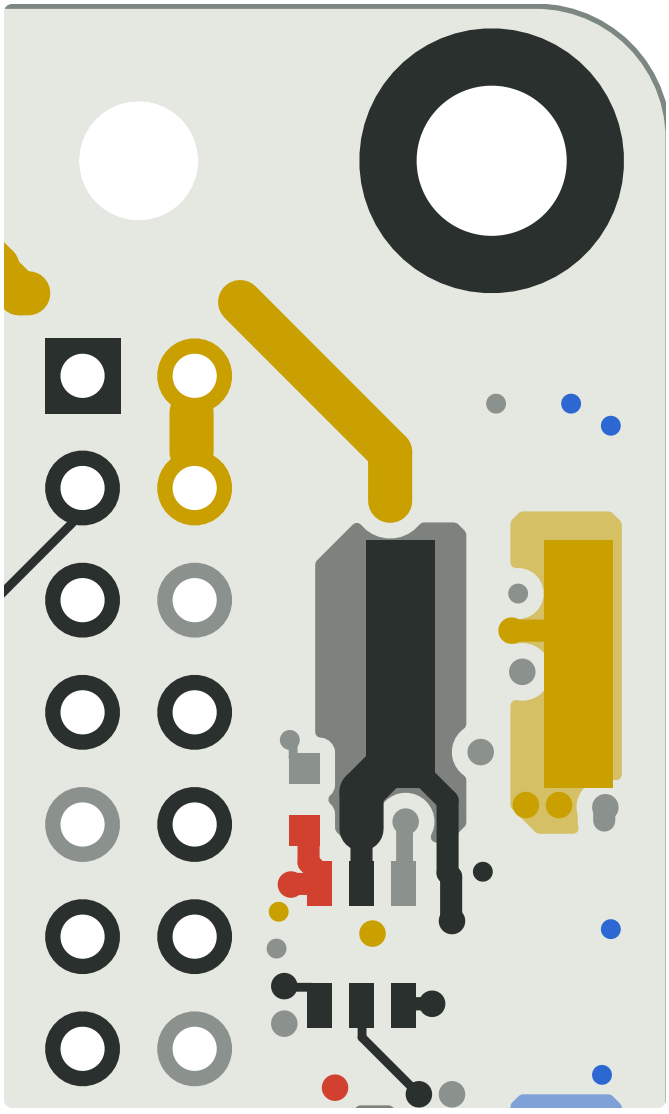
> **Fixed / as built:** JP1 left the strip's north end for the north band; the 5 V cell, the bottom 5V_PI pour and the Internal 2 patch are as session 2; the bottom GND island under C58 / TP38 is dropped by the fill now (island removal 'always', DFM row 24).

> **Versus the mote:** Sofar's 3V3 output sits on an In4 plane island with the same divider tap; here the same cell is copied twice.

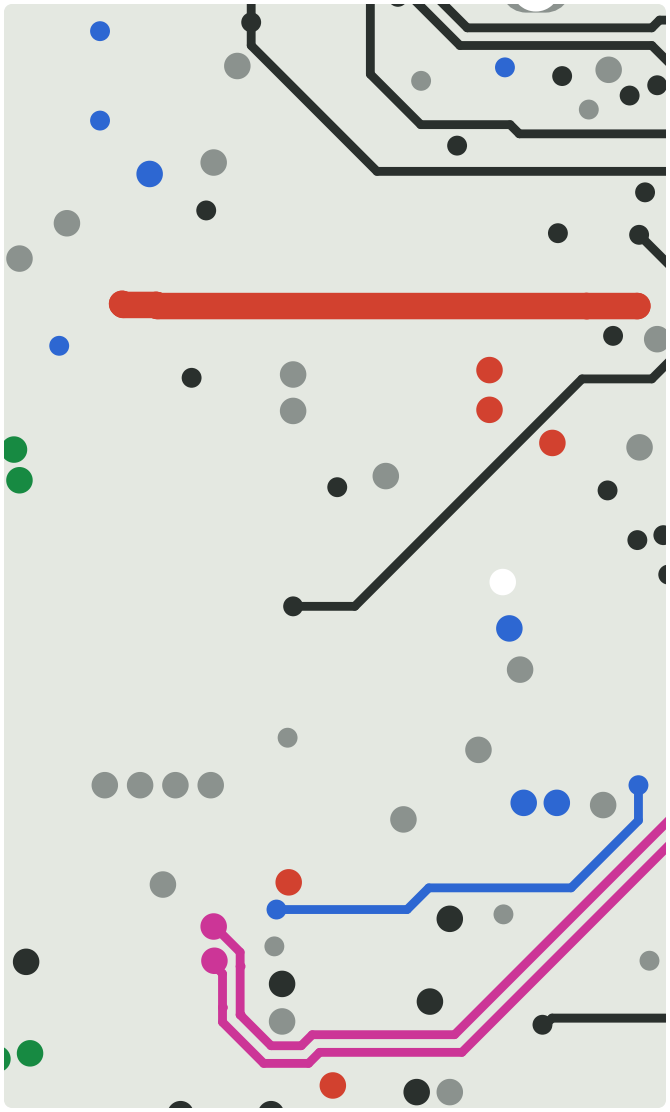
> **Still open:** None.



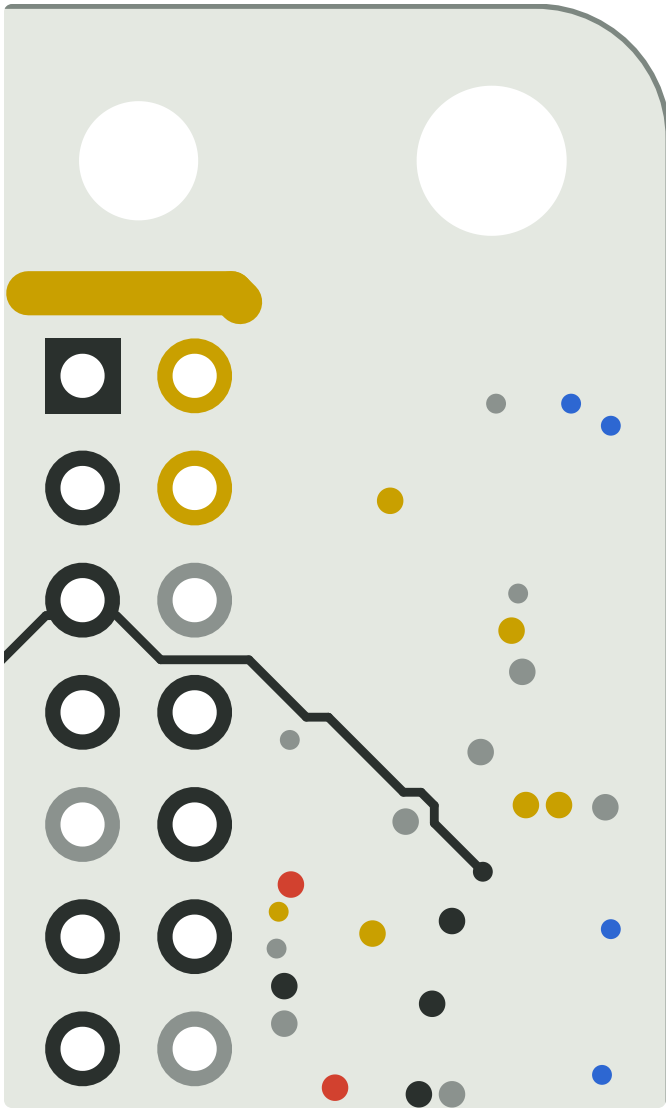
Sofar's mote, Top (block B5V transform)



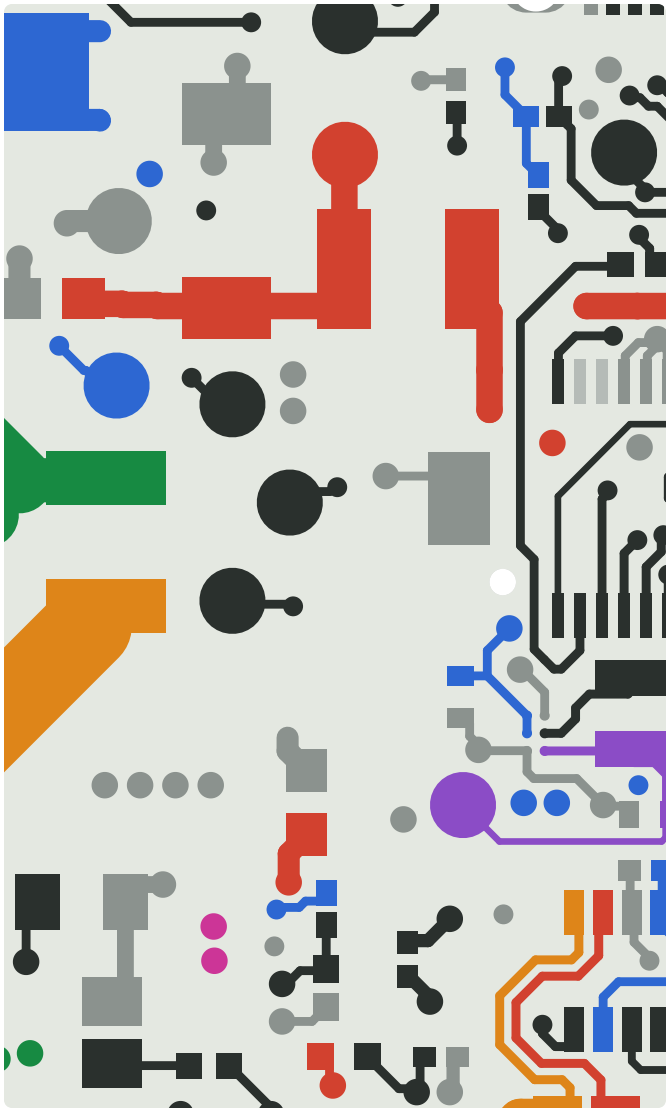
This board, Top



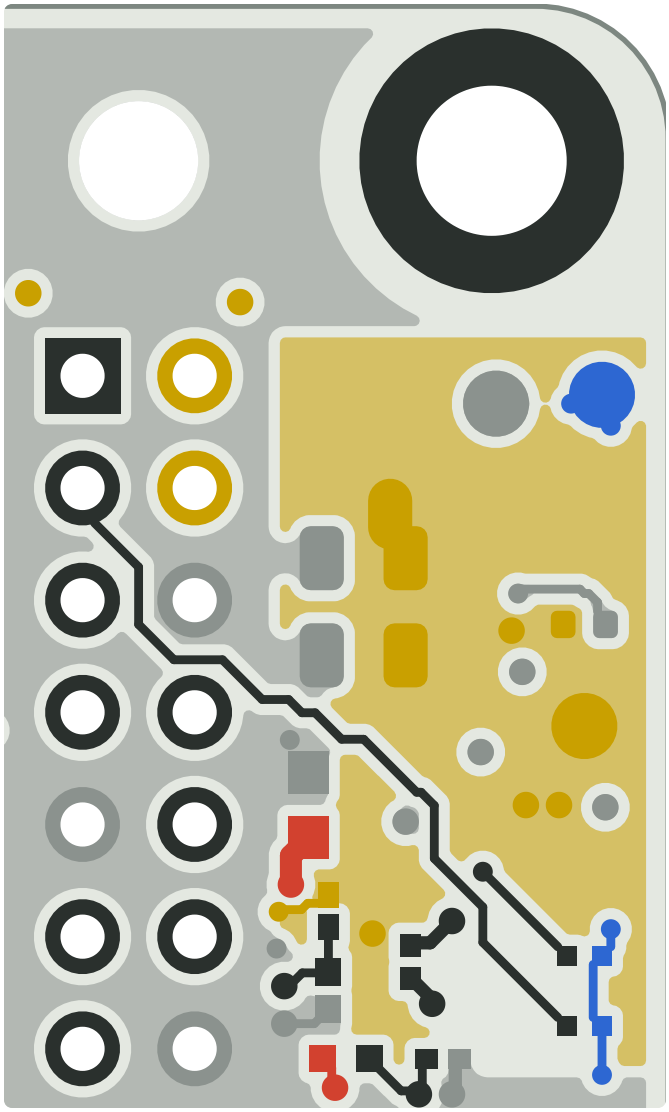
Sofar's mote, Internal 1 (block B5V transform)



This board, Internal 1



Sofar's mote, Bottom (block B5V transform)



This board, Bottom

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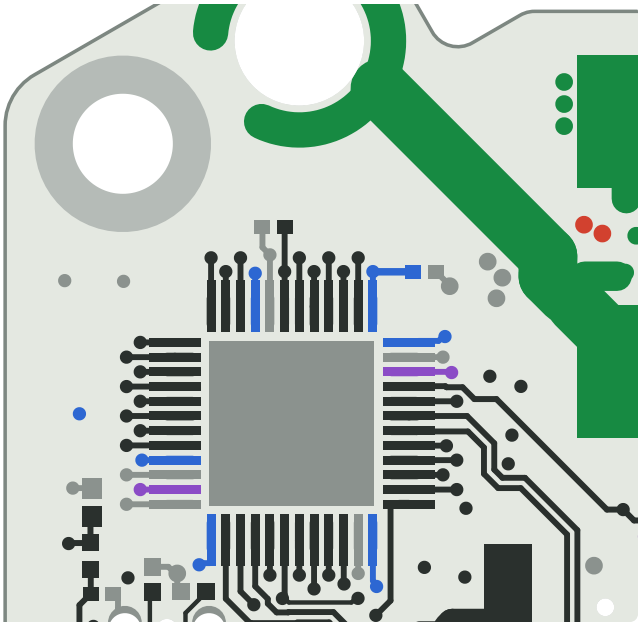
insert mp1 x -10.5…12.0, y 39.0…60.0 mm · block P1L

Port-1 inductor L1 (3 mm south) with the insert rings MP1 / MP2 (1.5 mm nearer the wall, 2.5 mm south) and the bus feeds; the P1L block's copper

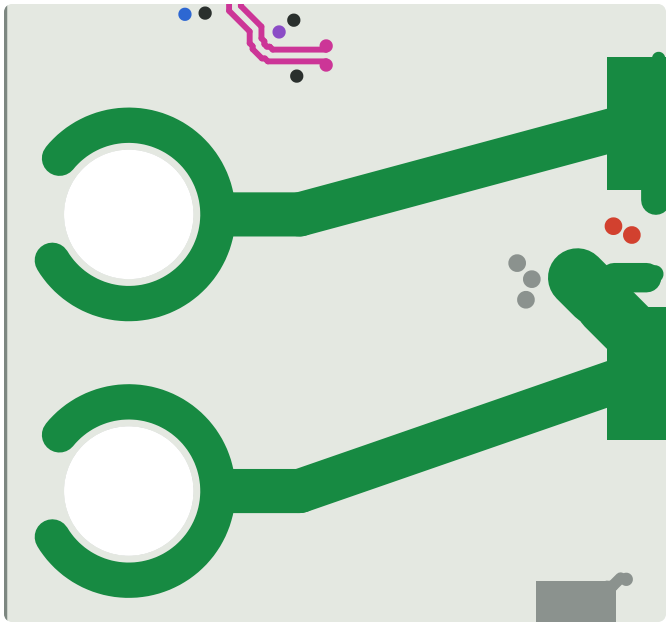
> **Fixed / as built:** The inserts 1.5 mm nearer the wall (ring copper on the brief's 0.5 mm edge line), the port-1 pair 2.5 mm south with L1's envelope (3 mm); the 1.5 mm bus feeds start 3.31 mm from the insert centre (NPTH-to-track, DFM row 13); the planes keep 0.3 mm from holes (DFM row 12).

> **Versus the mote:** Rigid copy of the inductor block; the feeds and the data legs are new (the mote's inserts are elsewhere).

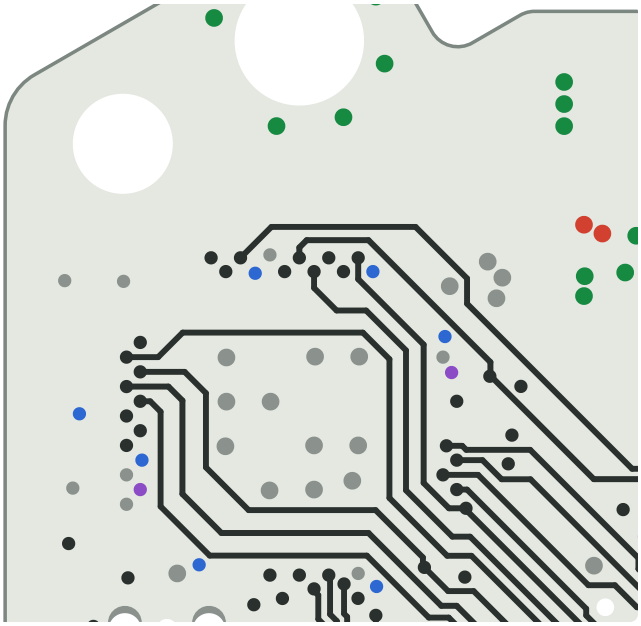
> **Still open:** Sofar's 2.0 mm BM1_N stub at L1 pad 2 is 0.31 mm from Sofar's GND via (the bus rule asks 0.35): 1 DRC item, declared. The insert contact's 76 DRC items are the mote's (Sofar Q6).



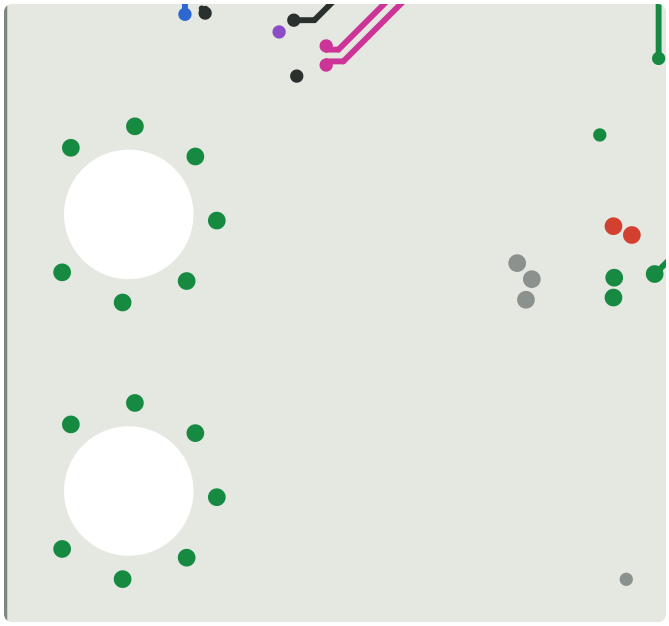
Sofar's mote, Top (block P1L transform)



This board, Top



Sofar's mote, Internal 1 (block P1L transform)



This board, Internal 1